AE 452 Soil and Water Engineering II (2,3,3)

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Infiltration and Soil Water Movement

Infiltration and Soil Water Movement

- **Infiltration** is the process of water entry into soil from precipitation or irrigation.
- **Soil water movement** is the process of water flow from one point to another within the soil.
- **Note**: The two processes cannot be easily separated as the rate of infiltration is controlled by the rate of soil water movement below the surface after the infiltration event, as the infiltrated water is redistributed.

Infiltration and Soil Water Movement

Importance: Infiltration and soil water movement play a key role in:

- surface runoff
- soil erosion
- groundwater recharge
- transport of chemicals in surface and subsurface water.

Infiltration and Soil Water Movement

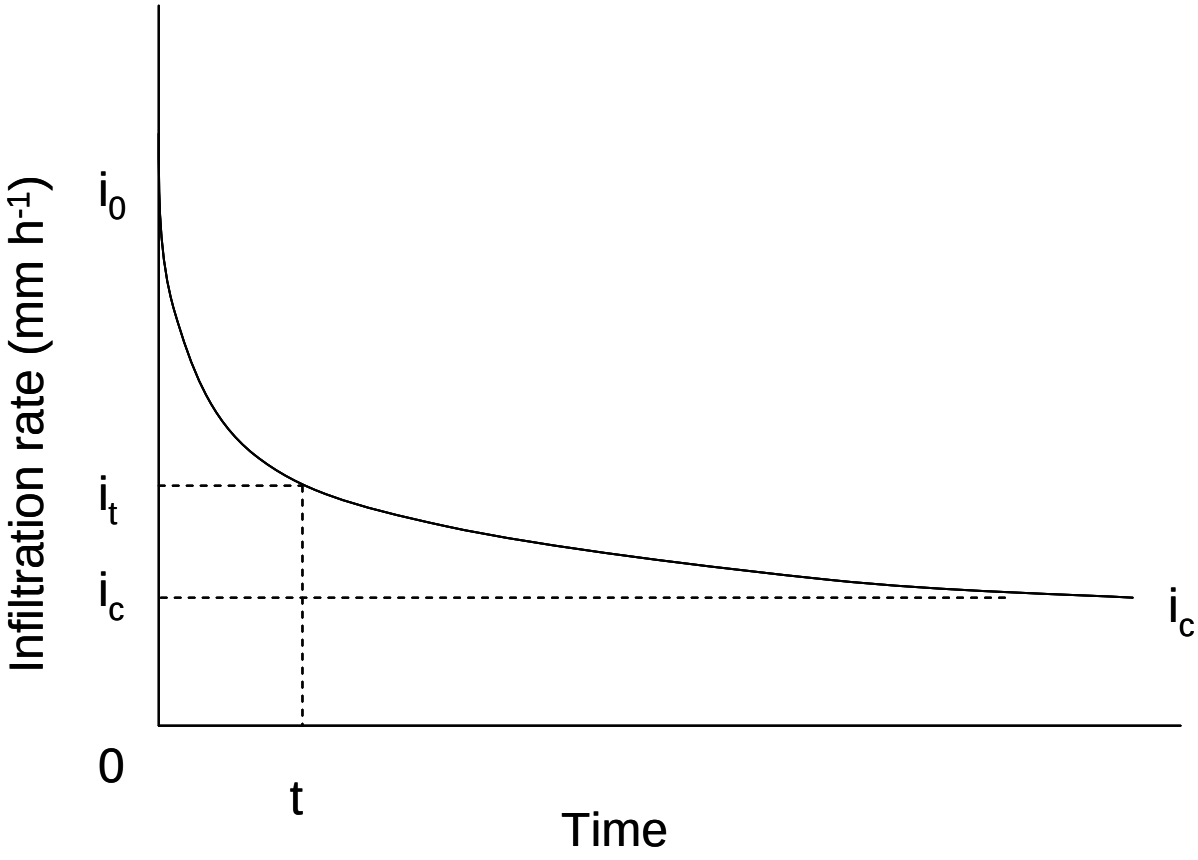


Fig. Infiltration rate with time

For a given soil, water infiltrates the soil at an initial rate (i_0) depending on the existing moisture content. The infiltration rate at any time, t (i_t) decreases until it reaches a final constant value referred to as infiltration capacity (i_c) when the soil is so wet that it is less able to take up water.

Empirical infiltration equations

Using i for the infiltrability as a volume flux (i.e. the volume of water entering a unit soil surface area per unit time) and I as cumulative volume of water infiltrated in time t per unit area of soil surface.

Then, $$i = dI/dt$$

Therefore, $$I = \int_0^t i\,dt$$

Empirical infiltration equations

Green and Ampt equation

$$i = i_c + b/I$$

Where i_c and b are the characterizing constants, with i_c being the infiltration capacity or the asymptotic steady infiltration flux attained when t becomes large. Note: at t = 0, I = 0, therefore i = infinite initially.

Empirical infiltration equations

Kostiakov equation

$$i = Bt^{-n}$$

Where B and n are constants. This implies at large t, i approaches zero rather than a non-zero constant i_c.

Empirical infiltration equations

Horton equation

$$i = i_c + (i_0 - i_c)e^{-kt}$$

Where i_c, i_0 and k are characterizing constants. At, t = 0, the infiltrability is not infinite but takes on the finite value i_0. This equation is more difficult to use because of the three constants that must be determine experimentally.

Empirical infiltration equations

Philips equation

$$i = i_c + \frac{s}{2t^{1/2}}$$

Where i_c and S are the characterizing constants. The i_c is a function of the transmission constant of hydraulic conductivity of the soil. S is the sorptivity.

Infiltration and Soil Water Movement

Factors influencing infiltration rate **(1/2)**:

1. Surface cover/condition: surface crust, compaction and presence of vegetation
2. Soil type - determines
   - hydraulic conductivity
   - water retention characteristics (i.e. the soils ability to store and release water)
   - physical properties
     - particle size distribution/texture
     - bulk density
     - organic matter
     - clay type







Soil Crust




Infiltration and Soil Water Movement

Factors influencing infiltration rate (2/2) :

3. Initial soil moisture content

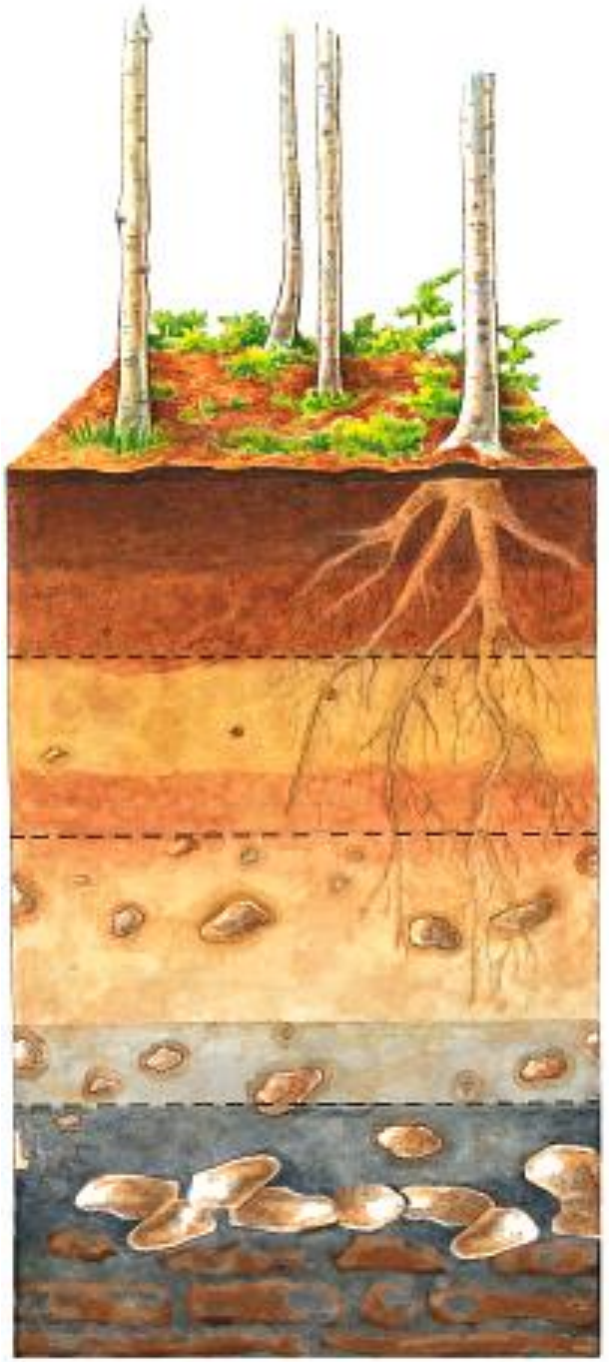
4. Time from the onset of water application (i.e. rain or irrigation)

5. Soil profile depth and layering

Soil Profile

- Soil Profile - A series of horizontal layers of different chemical composition, physical properties, e.g. particle size, and amount of organic matter observed when a vertical cross section is taken through the soil.

- Horizon - Each recognizable layer of the profile.



Measurement of infiltration rate

1. Cylinder infiltrometers: single or double ring infiltrometers
2. Disc infiltrometers
3. Sprinkling infiltrometers
4. Furrow method







Infiltration methods









Transport Processes in Soil

Flow of water in soils involves two distinct aspects that give rise to the water transport

- **Movement**: change of position of water with respect to a chosen reference
- **Accumulation**: increase or decrease of water at a given position with time

Transport Processes in Soil

Flux is the rate of movement of water in soil.

Flux density is the volume of water (V) flowing through a specified cross-section of soil perpendicular to the direction of flow during a certain time interval, divided by the area of the cross-section (A) and the magnitude of the time interval (t).

$$Flux\,(Q) \rightarrow \frac{Volume\ of\ \ water\,(V)}{Time\,(t)}$$

Transport Processes in Soil

Driving force: transport of water occurs when the static equilibrium is disturbed

$$Driving\ force = Gradient\ of\ hydraulic\ potential = -\frac{\delta H}{\delta z}$$

Darcy's Law

$$q = \frac{Q}{A \cdot t} = -K_s \frac{\delta H}{\delta z}$$

K_s is the saturated hydraulic conductivity

Note: The **minus or negative sign** indicates that the driving force and the ensuing flow of water are **directed** towards **decreasing values** of the hydraulic potential.

Groundwater

GROUNDWATER

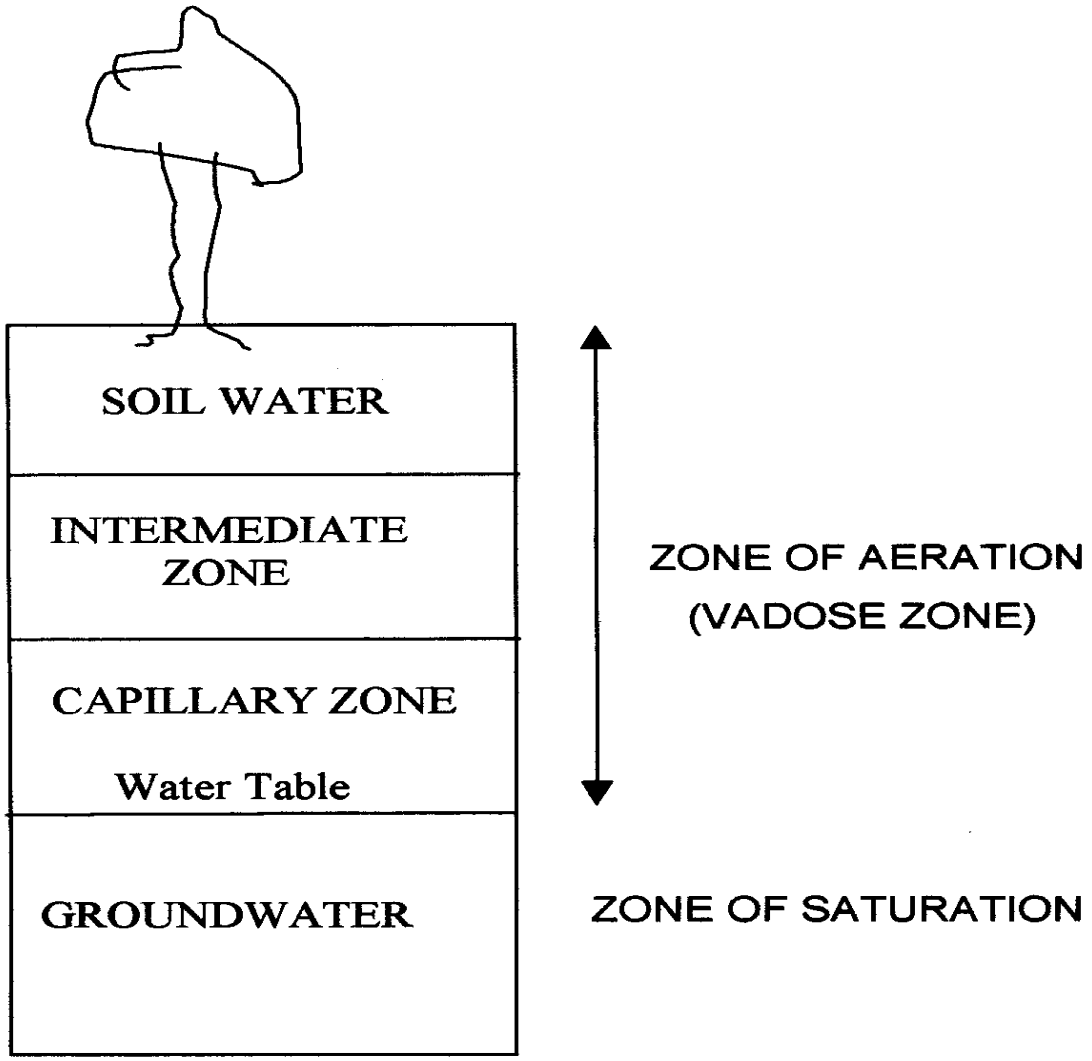


Fig.: Schematic illustration of Groundwater occurrence

GROUNDWATER

Zones of Aeration and Saturation

- **Zone of aeration**: zone that have interstices occupied partly by air and water
- **Capillary zone**: It extends from the water table (top of the groundwater) to the limit of capillary rise. The zone has variable thickness.
- **Zone of saturation**: zone with all interstices or pores filled with water under hydrostatic pressure i.e. the zone below the water table

GROUNDWATER

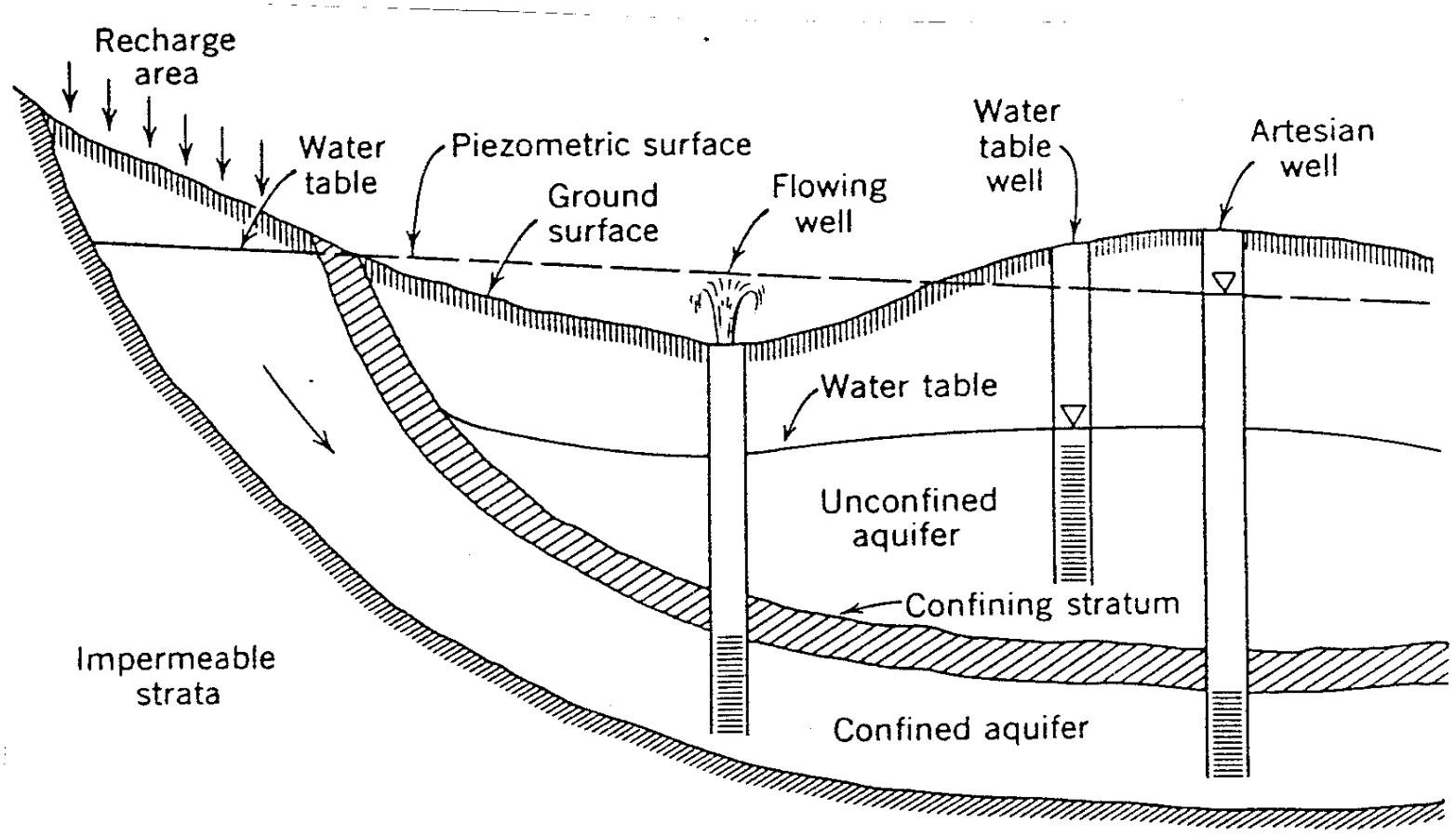


Fig.: Schematic cross section illustrating confined and unconfined aquifer

Groundwater: Terminology

Aquifer: A geologic formation or stratum containing water in its voids or pores that may be extracted economically and used as a source of water supply. An aquifer may be confined or unconfined.

- **Confined Aquifer**: One in which groundwater is confined under pressure greater than atmospheric by an overlying impermeable strata. It is also known as **artesian** or **pressure aquifer**.

- **Unconfined Aquifer:** One in which the water table serves as the upper surface of the zone of saturation. It is also known as a **free**, **phreatic** or **non-artesian aquifer**.

Groundwater: Terminology

- **Aquiclude:** A geologic formation so impervious that for all practical purposes it completely obstructs the flow of groundwater.
- **Aquitard:** A geologic formation which does not readily transmit water but stores water. It may transmit water very slowly vertically to or from a confined aquifer.
- **Spring**: Abrupt intersection of the ground by the groundwater table. Unlike swamps, the point of intersection is usually followed by a steep slope causing the water to flow by gravity.

Groundwater: Terminology

- **Porosity:** Groundwater occurs in voids, or pores of geologic formations. Porosity is defined as W/V. where W is the volume of water required to saturate all voids or volume of all pores and V is the total volume of the rock (soil).

- **Specific Retention (Sr):** When groundwater is pumped or drained, some water is retained by molecular and surface tension forces. Specific retention is therefore expressed as: Sr = Wr/V (usually expressed in %).

  Where Wr is the volume of water retained and V the total volume of rock (soil).

Groundwater: Terminology

- **Specific Yield (Sy):** The water removed by the force of gravity is the specific yield or effective porosity. It is defined as: $Sy = Wy/V$ (usually expressed in %).

  Wy is the volume of water drained while V is the total volume of rock(soil).

  Since total pore volume, $W = Wy + Wr$

  Porosity: $= Sy + Sr$

Groundwater: Terminology

- **Storage Coefficient (Storativity, S):** It is the volume of water which an aquifer releases from or takes into storage per unit surface area per unit change in piezometric head.

  For the unconfined aquifer, $S = Sr$

- **Transmissivity (T):** It is a measure of the ease of water movement in the rock (soil)

  It is the hydraulic conductivity (K) multiplied by aquifer thickness (b)

  i.e. $T = K \times b$

Humidity

Humidity

- **Atmospheric composition**: is a mixture of **dry air** (almost constant composition but with density decreasing with height) and **water vapour**
- **Absolute Humidity (AH):** Amount of water vapour in a unit volume of air (grams of water vapour/ cc of air). Vapour density (ρ_v) is usually referred to as Absolute humidity.

Humidity

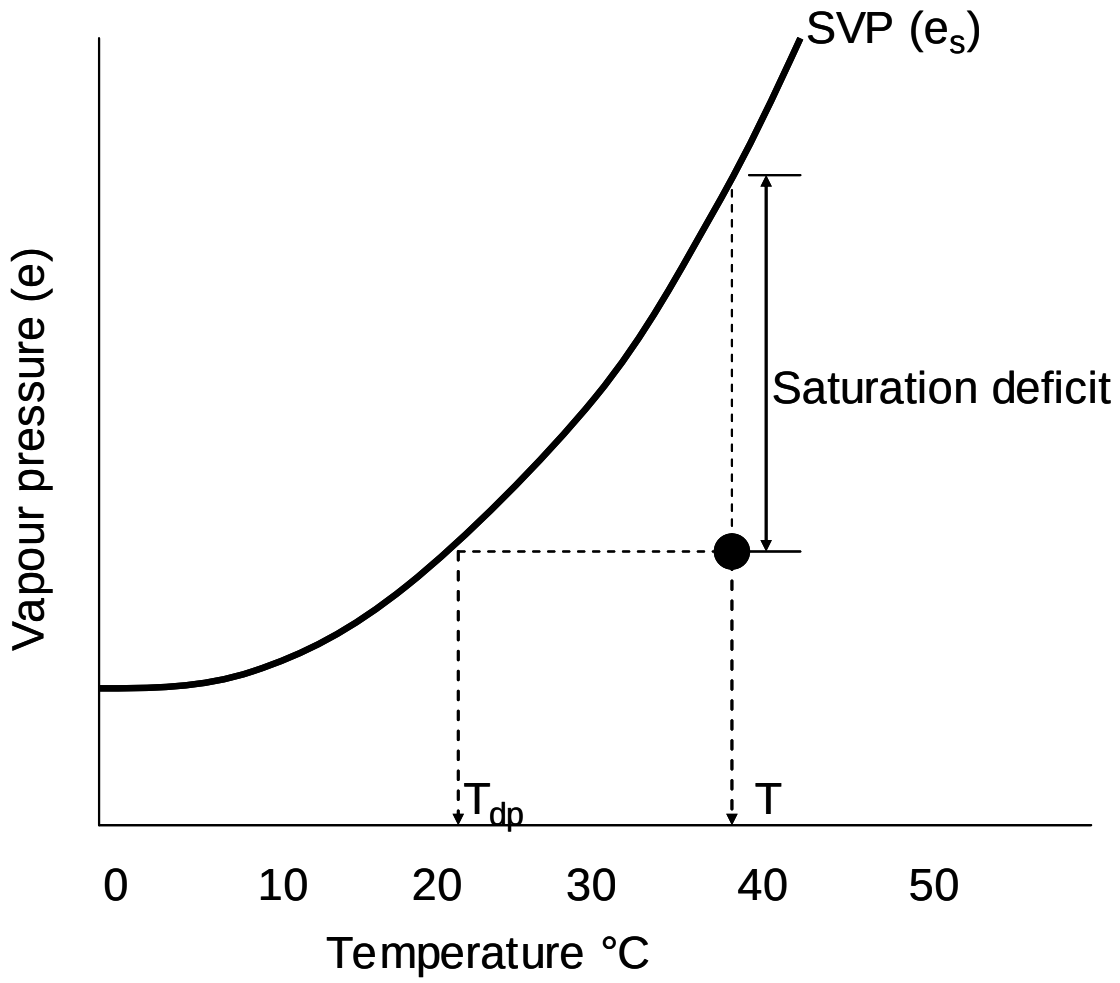
Water Vapour Capacity (WVC):

- Amount of water vapour that could be held in a given volume of air.
- WVC is dependent on temperature
- warm air can hold more moisture than cold air, implying warm has a higher WVC

Humidity

- **Saturation**: at constant temperature, an enclosed container partially filled with pure water at equilibrium, when the number of the water molecules leaving the liquid surface per second is the same as the number being captured per second then it is said to be saturated
- At saturation the vapour pressure is the maximum possible for that particular temperature under the conditions given which is referred to as saturated vapour pressure (svp or e_s).
    - $\mathrm{RH} = 100\%$
    - $\mathrm{AH} = \mathrm{WVC}$

Humidity



Saturation deficit: is when the vapour pressure (e) is less than svp (e_s) at the air temperature T and is given as the difference ($e_s - e$).

Figure : Vapour pressure of water as a function of temperature

Humidity

Relative Humidity:

Def.: Is the ratio of how much water the air has to the amount that it could have, expressed in percentage

- RH = (AH/WVC)x100
- RH = e/e_s
- Humidity is measured using a psychrometer. *Read on psychrometry

Evapotranspiration

Evapotranspiration

Evaporation: Is the transfer of water vapour into the atmosphere from evaporating surfaces.

Evaporation surfaces: plants, soil and water surfaces.

Transpiration: is the transfer of water which has been transmitted from the soil through the plant from plant surfaces.

Evapotranspiration

Evapotranspiration: Is the transfer of water from moist surfaces and living plant into the atmosphere (i.e. evaporation + transpiration)

Evaporation power: Is the ability of air to transform a mass of liquid water into vapour and diffuse it.

Evapotranspiration

Conditions/requirement under which evapotranspiration occurs:

1. **Energy**: Supply of heat to provide the latent heat of vaporization
2. **Vapour pressure difference/gradient**: Vapour pressure in the overlying air must be maintained at less than that at the evaporation surface
3. **Water source**: Water must continuously be available for evaporation

Estimating evapotranspiration

Three (3) groups of methods-

1. **Mass transfer**: It is assumed that water moves away from evaporating and transpiring surfaces in response to the combined phenomenon of turbulent mixing of the air and vapour pressure gradient. This approach requires the measurement of wind velocity and humidity at two elevations and it is rarely practical.

Estimating evapotranspiration

2. **Energy balance**: Heat is required for evaporation, therefore if there is no change in water temperature then the net radiation or heat supplied is a measure of evaporation. This approach is more practical. Eg. Penman equation
3. **Empirical methods**: This is based on the assumption that the energy available for evaporation is proportional to the temperature. Eg. Blaney-Criddle

Pan Evaporation

Class A pan

Characteristics

- Material type - galvanized pan
- Material thickness - 0.8 mm
- Pan diameter - 1.21 m
- Pan depth - 25.5 cm
- Pan colour - grey

- **Placement** - mounted on a wooden stand (15 cm above ground level) such that air freely circulates around and under the pan placed in an open space
- **Measurement** - A ruler or a micrometre screw gauge in a stilling well is used to measure the drop in water.

Pan Evaporation Calculation

Day/Date	Reading from pan (one reading every 3 days	Rain (mm)	Evaporation power of the average reading per 3 day-period
Tue.			
Wed.			
Thurs.	27		27/3=9.0mm/day
Fri.			
Sat.		9	
Sun.	25		(25+9)/3= 11.3mm/day
Mon			
Tue.			
Wed.	29		29/3=9.7mm/day

Total evaporation for the 9 days: 81mm + rain
Average daily Evaporation: (81+9)/9 =10.0 mm/day

Pan Evaporation Calculation

Evaporation on rainy days: The rainfall depth is added to the reading on the pan gauge.

Example: If the daily pan reading indicates 6mm and the rainfall recorded for the same day is 8mm then the evaporation for that day is 14mm

Factors that influences evapotranspiration (1/2)

1. Degree of soil water saturation
2. Soil texture and structure, which influences capillary rise and water retention
3. Net radiation
4. Soil temperature
5. Air temperature surrounding the soil surface and the leaves
6. Saturation deficit

Factors that influences evapotranspiration (2/2)

7. Wind speed
8. Roughness of evaporating surface
9. Level of root and leaf development, as it determines the capacity of the plant to take up water for transpiration
10. Salt concentration of the water as evapotranspiration decreases with increase in salt concentration
11. Elevation: Evaporation increases with elevation

Terminologies (1/2)

- **Potential evapotranspiration**: Is the maximum possible amount of evaporation and transpiration that takes place given unlimited supply of moisture under a given meteorological condition.

  That is the amount of water returned to the air by cultivated and uncultivated plants in optimal growing and watering condition.

- **Actual evapotranspiration**: when water supply is not adequate.

Terminologies (2/2)

Reference evapotranspiration (ET_o): is the potential evapotranspiration (ET) from a grass reference surface e.g. alfalfa. It is the maximum evapotranspiration of:

a. an extensive green grass cover
b. uniform height (8 -15 cm)
c. completely shading the ground
d. actively growing
e. no shortage of water

Methods of determining reference evapotranspiration (ET_o)

- Blaney-Criddle
- Penman ETo according to Frere and Popov
- Doorenbos and Pruitt
- Penman-Monteith